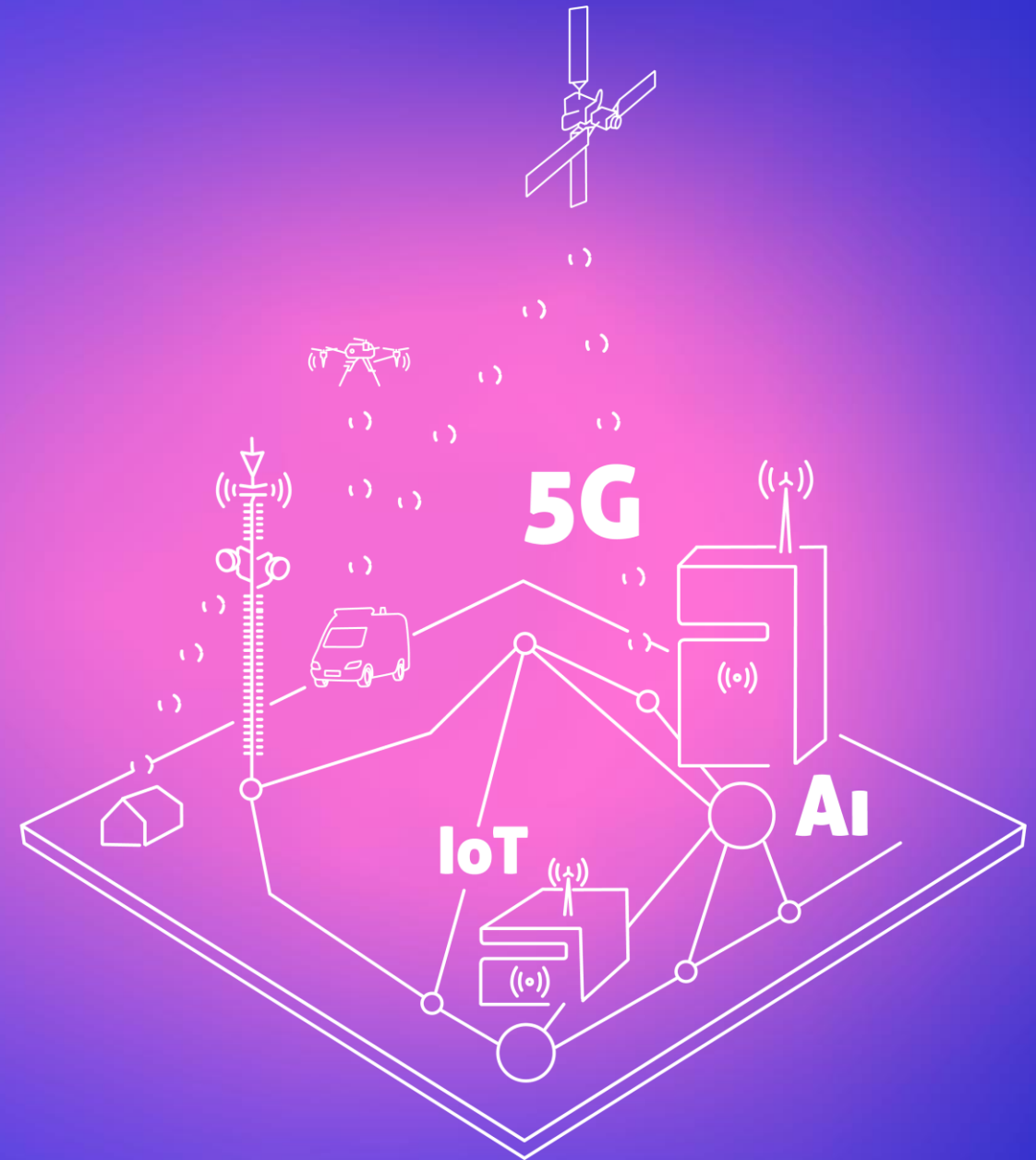




## Introduction: Data types

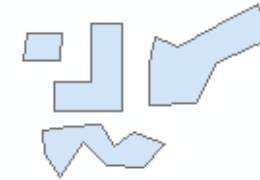
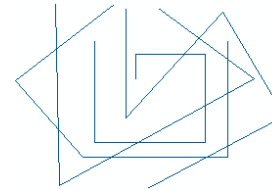
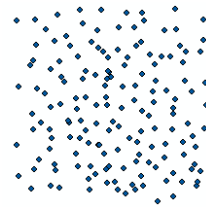
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## Data can be:

### 1. Vector

- Points
- Lines
- Polygons



### 2. Raster

- GeoTIFF



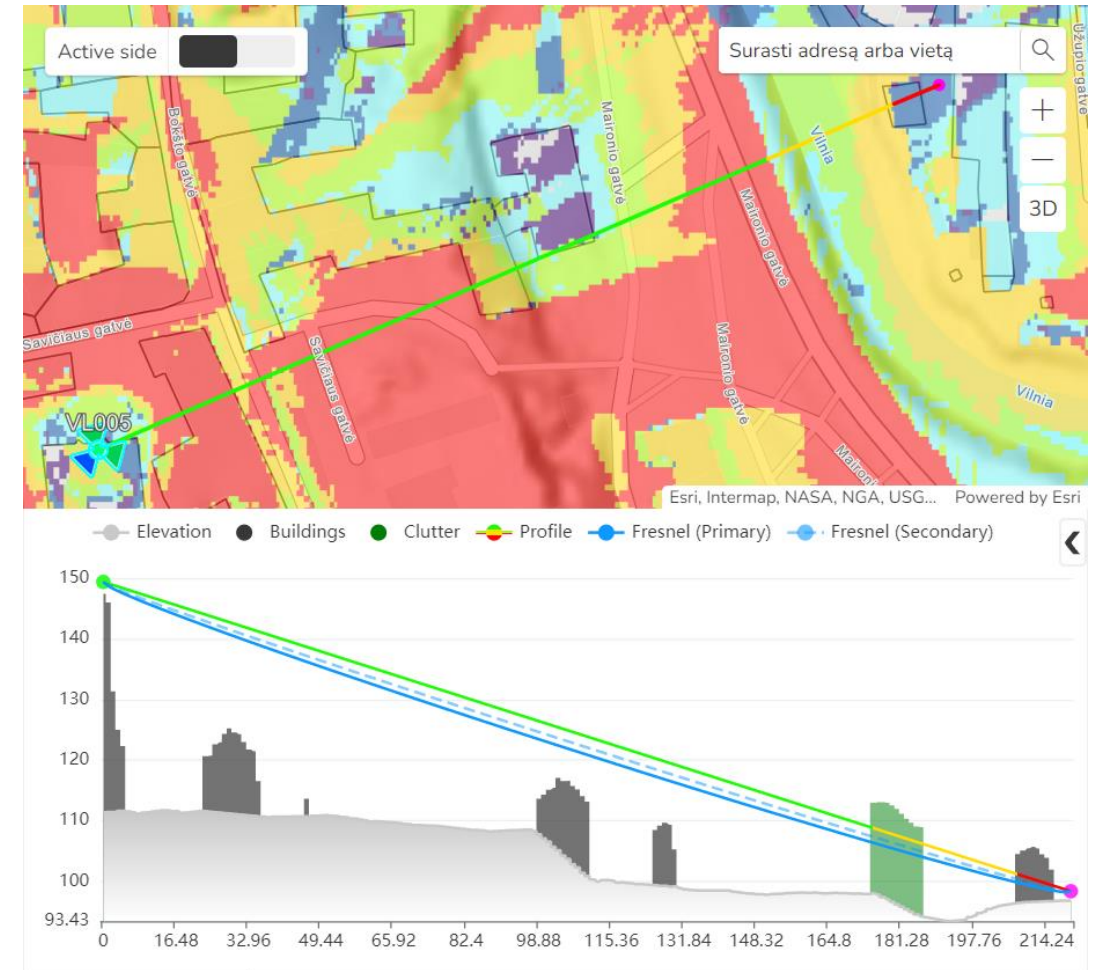
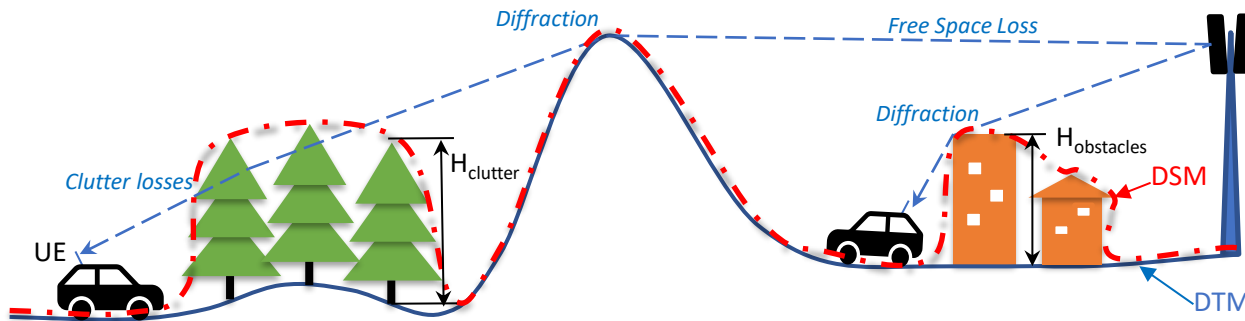
### 3. Tabular

| Height | Azimuth | Tilt | Frequency | Power | Antenna gain | Misc loss | Bandwidth | Noise figure | Downlink duplex factor | Subcarrier spacing |
|--------|---------|------|-----------|-------|--------------|-----------|-----------|--------------|------------------------|--------------------|
| 22     | 70      | 0    | 3500      | 65    | <Null>       | 0         | 40        | 6            | 0.6                    | 30                 |
| 24     | 190     | 0    | 3500      | 65    | <Null>       | 0         | 40        | 6            | 1                      | 30                 |
| 22     | 310     | 0    | 3500      | 65    | <Null>       | 0         | 40        | 6            | 1                      | 30                 |
| 30     | 60      | 0    | 3500      | 65    | <Null>       | 0         | 40        | 6            | 1                      | 30                 |
| 33     | 205     | 1    | 3500      | 65    | <Null>       | 0         | 40        | 6            | 1                      | 30                 |

# Modelling Outdoor coverage

The CE tools make use of three distinct GIS data layers to obtain high precision modelling of radio wave propagation losses:

1. **Digital Terrain Model (DTM)**, also known as Digital Elevation Model (DEM), which describes Earth surface, i.e., path terrain profile in terms of ground elevation above uniform sea level.
2. **Obstacles layer**, delineating buildings and other such objects above Earth surface that may be considered to be principal impediments for radio wave propagation.
3. **Clutter layer**, delineating natural occurring or human cultivated ground cover that may be partially penetrable by radio waves, such as natural vegetation (e.g., forests, trees, bushes) or various crops, gardens, parks, etc.



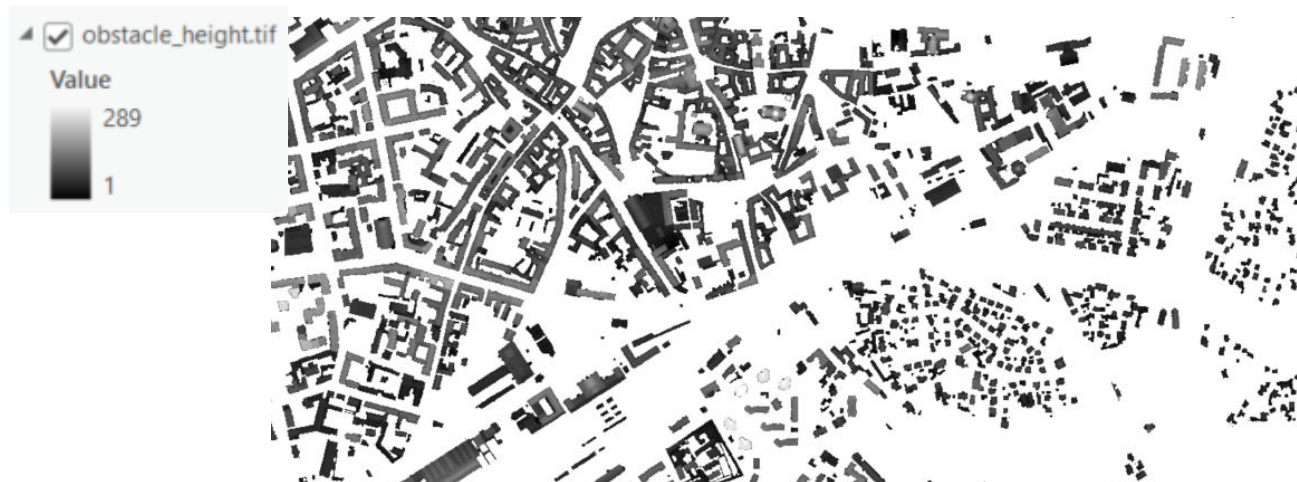
# Raster Type Input: Elevation

- Digital terrain model (DTM)
  - Represents Earth's ground/water level above sea level
  - GeoTIFF raster format
  - Height values in meters
  - Coordinate system – projected
  - Resolution (cell size) – centimeter level
  - Raster name: elevation.tif

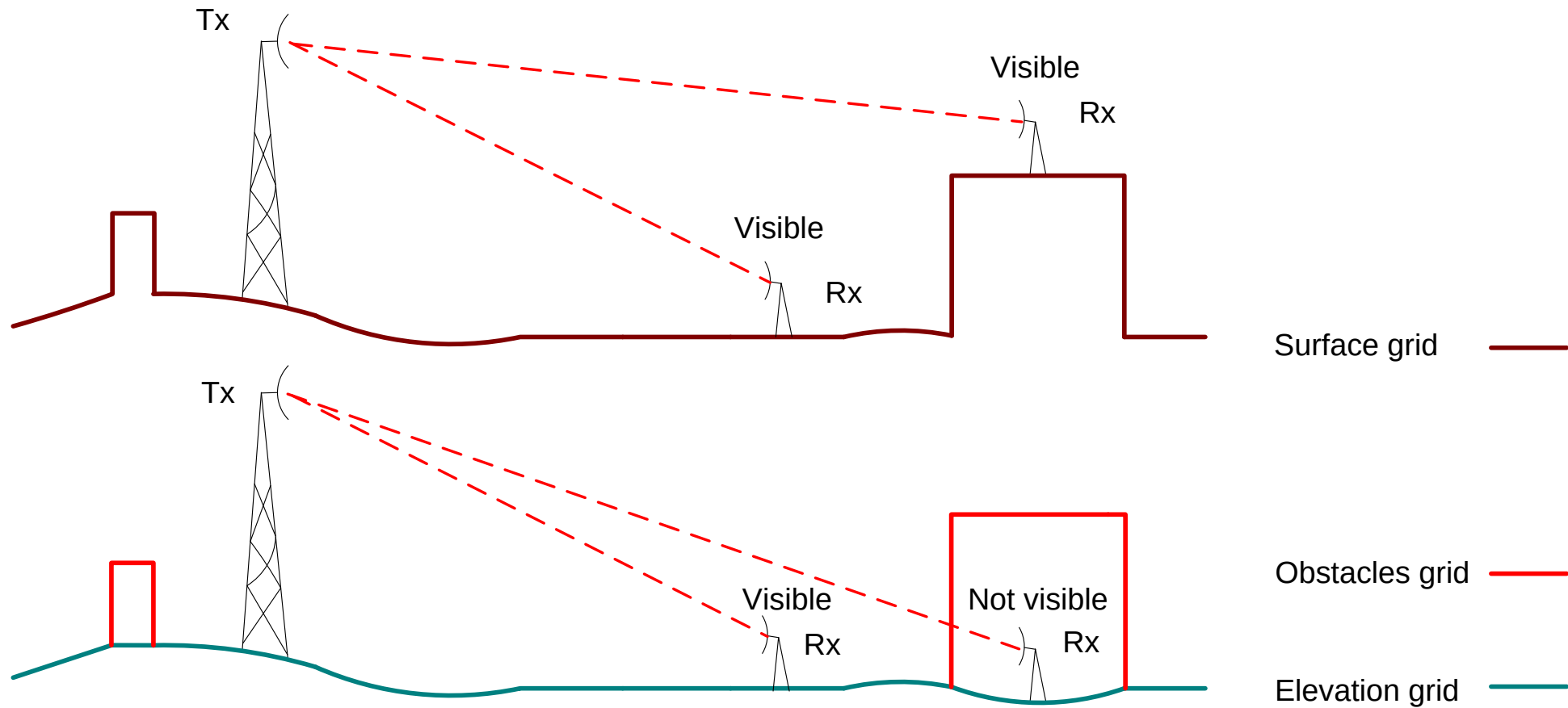


# Raster Type Input: Clutter height

- Clutter height
  - Represents objects height above elevation raster.
  - GeoTIFF raster format
  - Height values in meters
  - Coordinate system – projected
  - Resolution (cell size) – centimeter level
  - Raster name: clutterHeight.tif

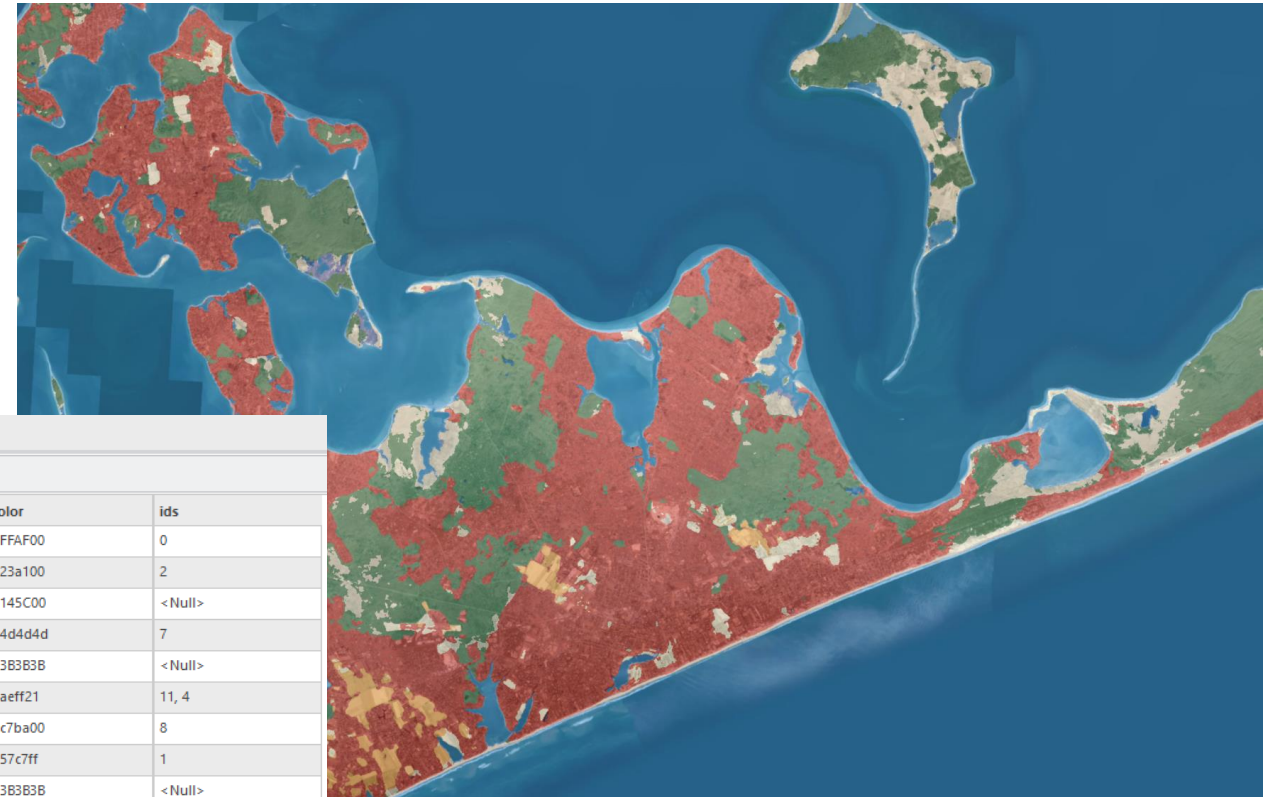


# DTM vs. Surface



# Clutter classes

- Clutter classes
  - Represents land use classes.
  - GeoTIFF raster format
  - Coordinate system – projected
  - Resolution (cell size) – centimeter level
  - Raster name: clutterClasses.tif



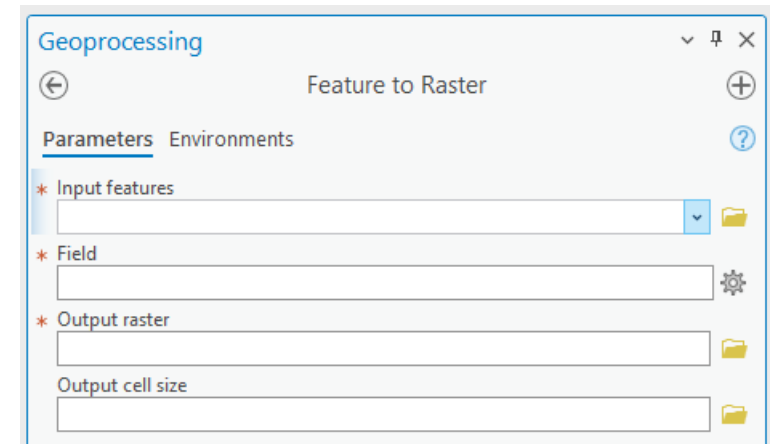
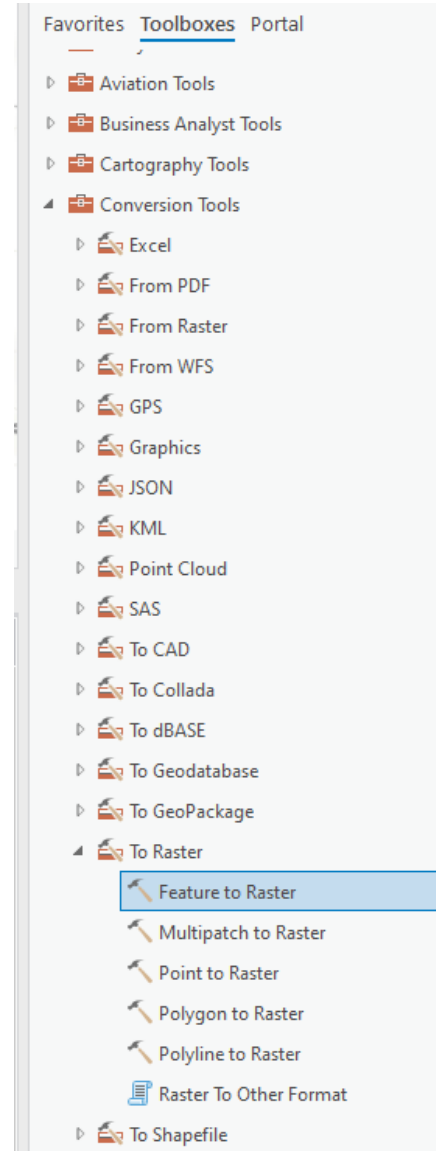
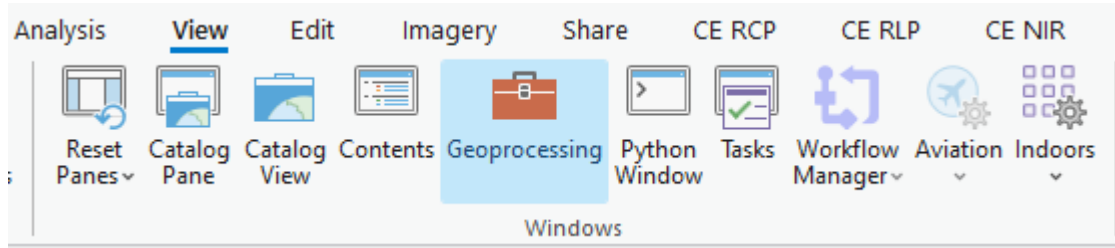
| clutter                                                                                            |                    |                  |        |                  |                      |                       |                      |            |         |        |
|----------------------------------------------------------------------------------------------------|--------------------|------------------|--------|------------------|----------------------|-----------------------|----------------------|------------|---------|--------|
| Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy Rows: Insert |                    |                  |        |                  |                      |                       |                      |            |         |        |
| OBJECTID *                                                                                         | clutter_class_name | clutter_class_id | height | nominal_distance | surface_refractivity | relative_permittivity | surface_conductivity | express_id | color   | ids    |
| 1                                                                                                  | buildings          | 1                | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #FFAF00 | 0      |
| 2                                                                                                  | forest             | 2                | 10     | 27               | 315                  | 13                    | 0.004                | 36         | #23a100 | 2      |
| 3                                                                                                  | denseForest        | 4                | 15     | 27               | 315                  | 13                    | 0.004                | 36         | #145C00 | <Null> |
| 4                                                                                                  | urban              | 5                | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #4d4d4d | 7      |
| 5                                                                                                  | denseUrban         | 7                | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #3B3B3B | <Null> |
| 6                                                                                                  | bareGround         | 8                | 0      | 27               | 315                  | 10                    | 0.002                | 36         | #aef221 | 11, 4  |
| 7                                                                                                  | crops              | 9                | 2      | 27               | 315                  | 20                    | 0.03                 | 36         | #c7ba00 | 8      |
| 8                                                                                                  | water              | 11               | 0      | 27               | 365                  | 80                    | 1                    | 36         | #57c7ff | 1      |
| 9                                                                                                  | road               | <Null>           | 0      | 27               | 315                  | 10                    | 0.002                | 36         | #3B3B3B | <Null> |
| Click to add new row.                                                                              |                    |                  |        |                  |                      |                       |                      |            |         |        |

# Clutter types

## Corine Land Cover

|     |                                       |
|-----|---------------------------------------|
| 111 | dichte Verbauung                      |
| 112 | lockere Verbauung                     |
| 121 | Industrie u. Gewerbe                  |
| 122 | Verkehrsfläche                        |
| 123 | Hafengebiet                           |
| 124 | Flughafen                             |
| 131 | Abbaufäche                            |
| 132 | Deponien, Abraumhalden                |
| 133 | Baustellen                            |
| 141 | Städtische Grünfläche                 |
| 142 | Freizeitanlage                        |
| 211 | Ackerland                             |
| 221 | Weinbau                               |
| 222 | Obstbestände                          |
| 231 | Wiese und Weide                       |
| 242 | Landwirtschaft komplex                |
| 243 | Landwirtschaft, natürliche Vegetation |
| 244 | Land/forstwirtschaftliche Flächen     |
| 311 | Laubwald                              |
| 312 | Nadelwald                             |
| 313 | Mischwald                             |
| 321 | Natürliches Grünland (alpine Matten)  |
| 322 | Latschen                              |
| 324 | Wald-Strauch Übergang                 |
| 331 | Strand, Sand                          |

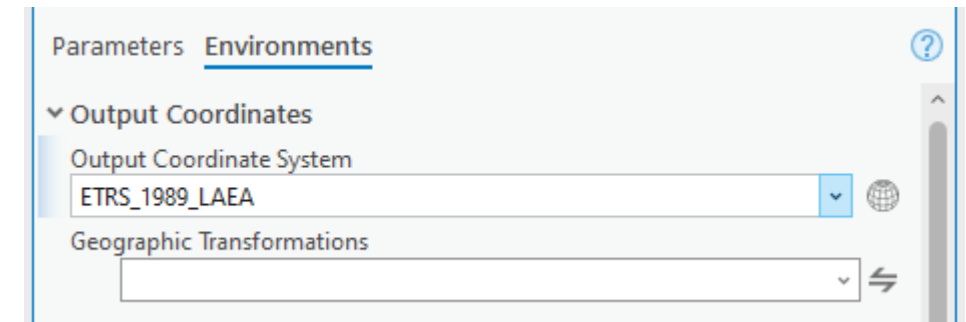
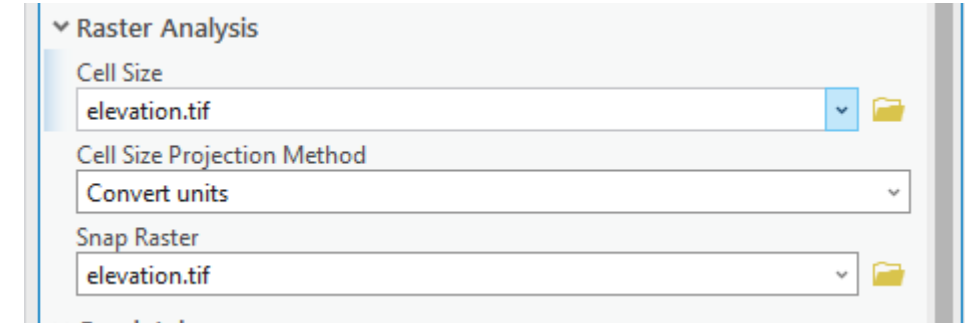
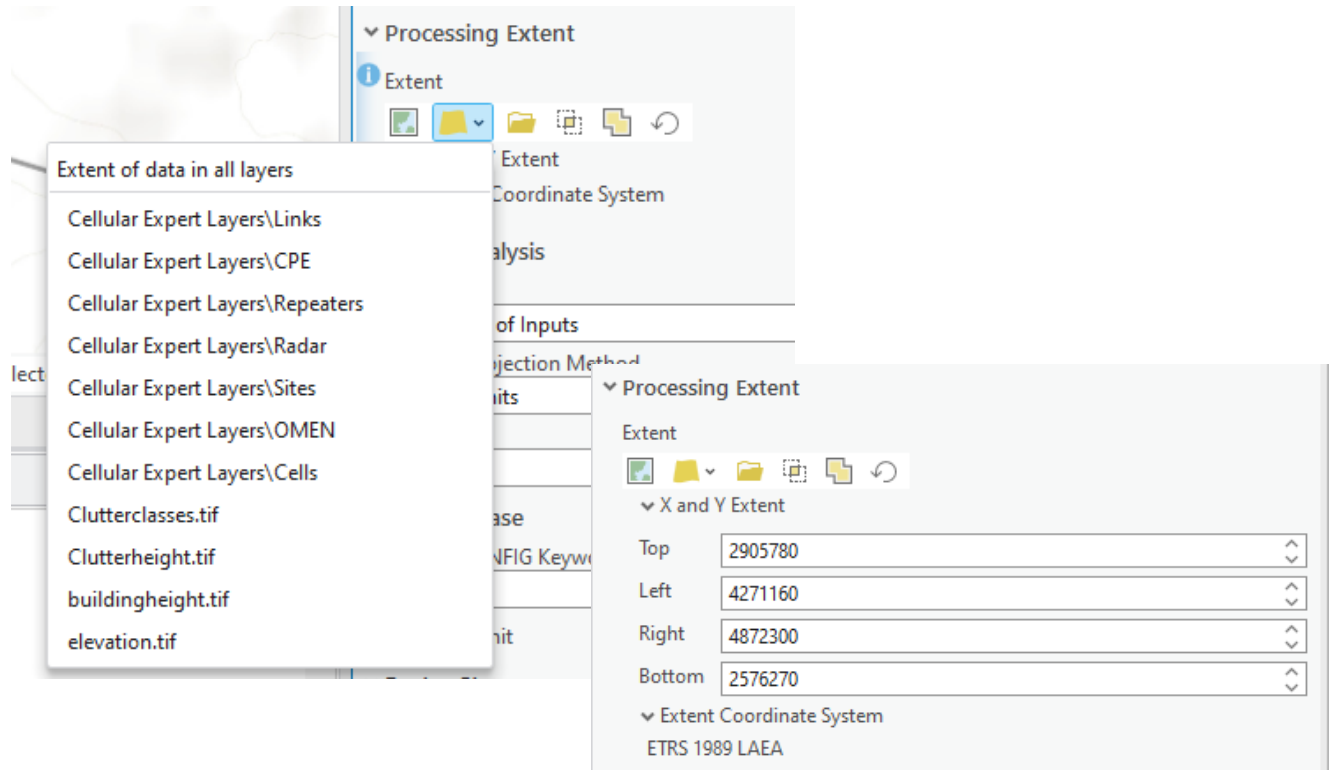
# Raster from vector data



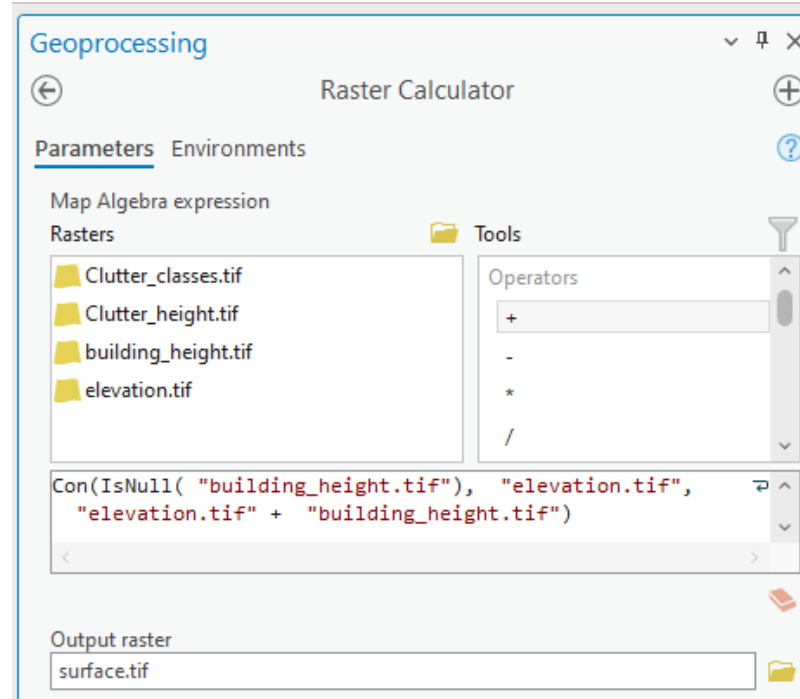
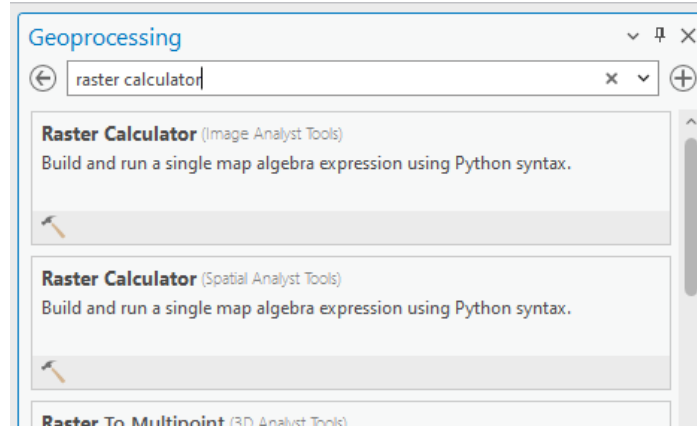
Following should be defined: input layer, output raster cell size, data field that will be converted to grid, output raster file name.

**Note:** if some features in input layer are selected, only selected ones will be converted.

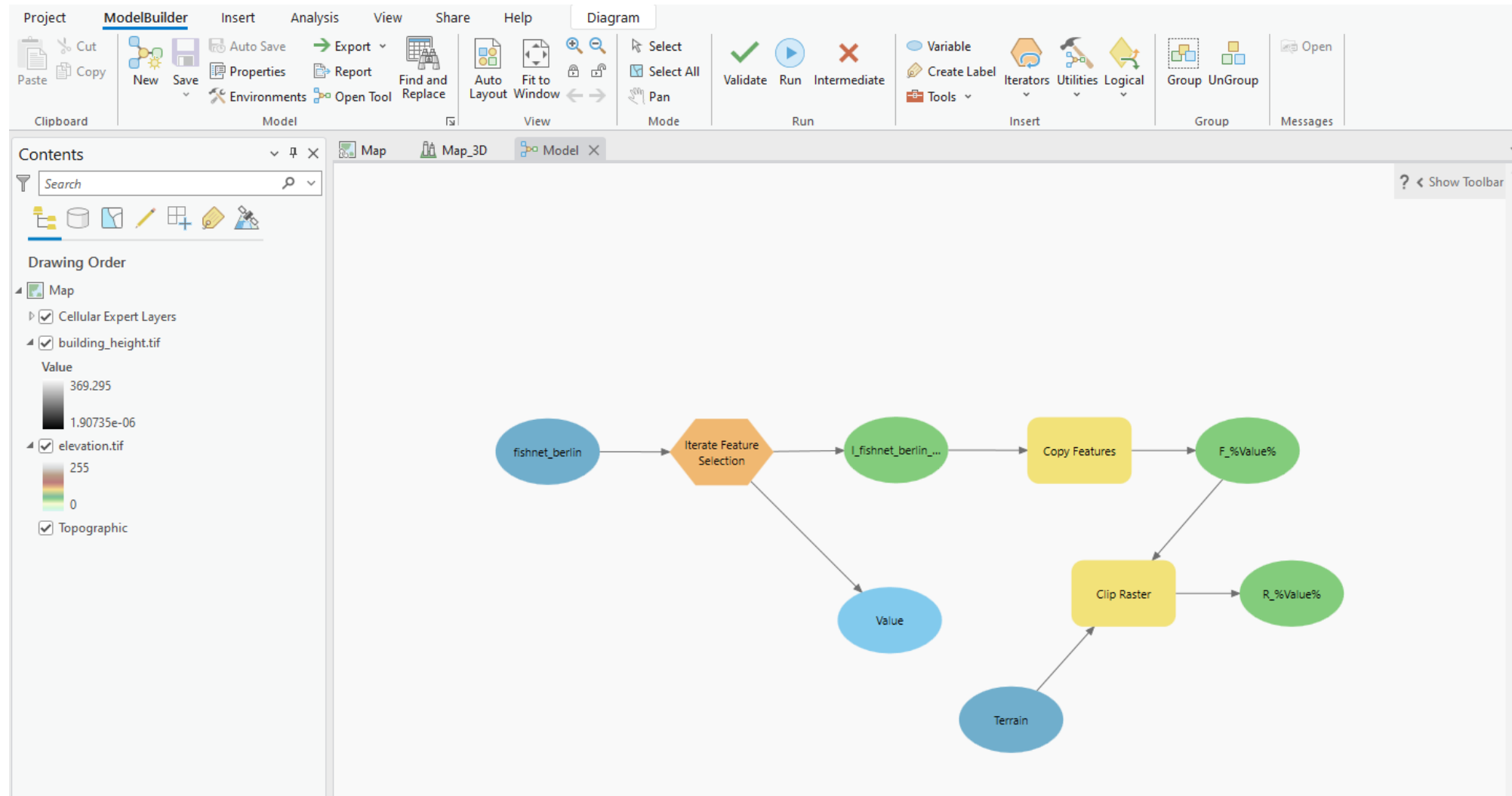
# Environment Settings



# Raster Calculator



## Automate your GIS tasks





Questions?

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